

Dr Puneeth V

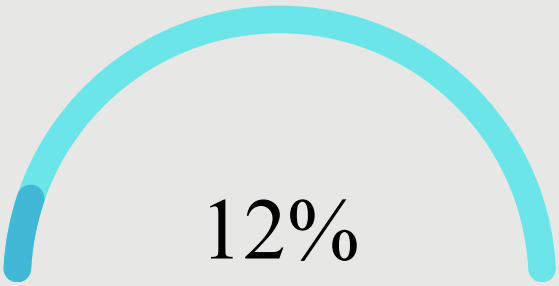
Assistant Professor, Department of Mathematics,
CHRIST (Deemed to be University)

Research-focused mathematician with strong expertise in IQAC documentation, learner-centric pedagogy, international research visibility, and academic governance, committed to building research-intensive and quality-driven academic ecosystems

+91-9663501058
www.christuniversity.in
puneeth.v@christunivesity.in
CHRIST (Deemed to be University), Bangalore 560029, India

EDUCATION

- **Specialization in Generative AI for Educators**
  IBM
  2025
- **PhD in Mathematics**
  CHRIST (Deemed to be University)
  2019 - 2023
- **MSc in Mathematics**
  CHRIST (Deemed to be University)
  2016 - 2018
- **BSc in Physics, Mathematics & Electronics**
  Christ University
  2013 - 2016
- **PUC in Physics, Chemistry, Mathematics & Biology**
  Christ Junior College
  2011 - 2013
- **SSLC**
  Lady Vailankanni High School
  2010-2011



12%

Service completed

MEMBERSHIPS

Member of the **International Association of Mathematical Physics**

PROFILES

GOOGLE SCHOLAR | SCOPUS | ORCID | RESEARCH GATE | PERSONAL HOME PAGE | OFFICIAL HOMEPAGE

SKILLS

Management Skills, Online Teaching, Math Softwares, Communication, Critical Thinking, Leadership

ABOUT ME

I am a research-driven mathematician, academic administrator, and educator who consciously aligns scholarly work, pedagogy, and governance with evolving academic and technological landscapes. I actively keep my research profile current by engaging with contemporary developments such as hybrid modelling frameworks that combine classical governing equations with machine learning approaches, including Physics-Informed Neural Networks and neural-network-based surrogate models. My sustained research output, editorial responsibilities, international collaborations, and recognition in Stanford University's Top Two Percent of the World's Most Influential Scientists reflect a commitment to continuous scholarly updating and global academic engagement rather than static specialization.

Beyond research, I remain professionally relevant by actively participating in academic governance and quality assurance processes. As IQAC Coordinator and Strategic Plan Coordinator, I work closely with institutional benchmarks, accreditation frameworks, outcome-based education metrics, and data-driven documentation practices. These roles require continuous learning, policy awareness, and adaptability to evolving higher-education regulations, ensuring that my administrative outlook remains current, compliant, and forward-looking.

My pedagogical practice is equally dynamic and learner-centric. With formal specialization in Generative AI for Educators, I consciously update my teaching methods by integrating educational technology, AI-assisted learning tools, and computational platforms into the classroom. I use AI and digital tools not as static aids, but as evolving pedagogical partners that support personalized learning, formative assessment, and critical thinking, while maintaining ethical and academic rigor.

I view relevance as an ongoing process rather than a fixed achievement maintained through research engagement, interdisciplinary exploration, administrative responsibility, and reflective teaching practice. This approach enables me to contribute meaningfully to research-intensive institutions by staying intellectually current, technologically aware, and academically agile.

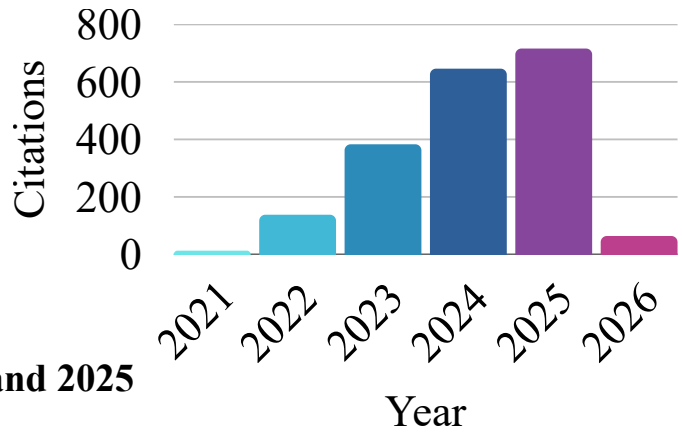
PERSONAL DEATAILS

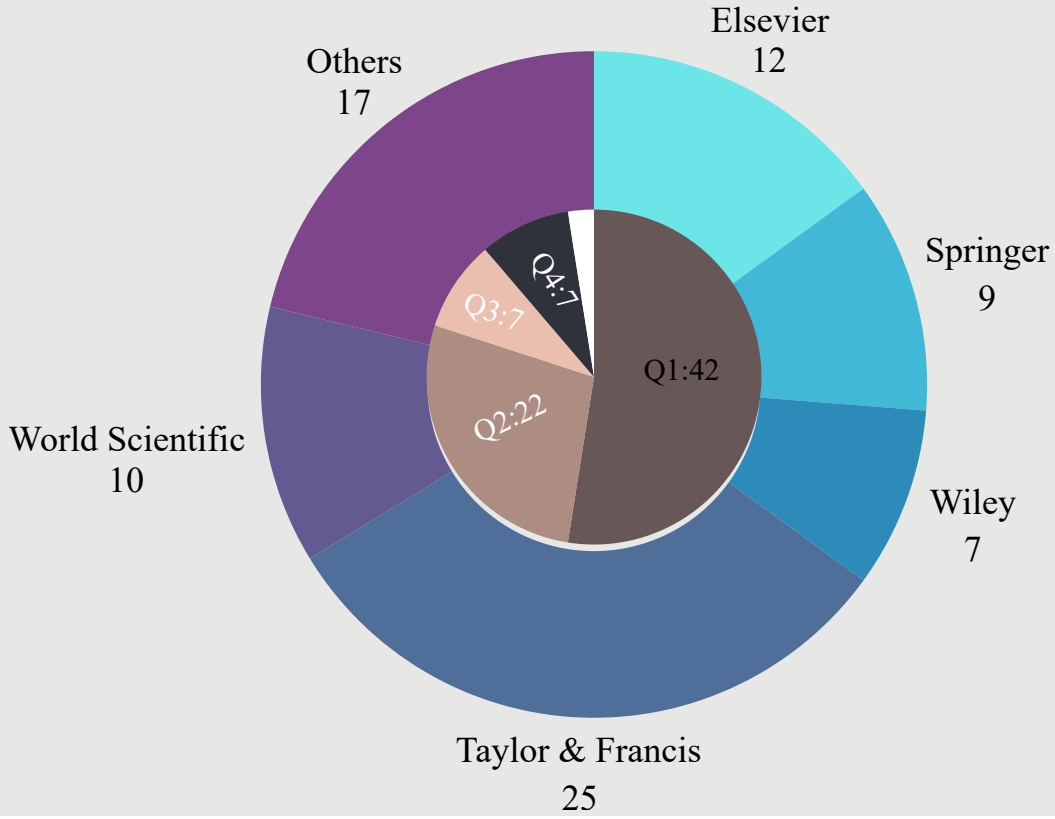
DOB: 11 May 1995
Father: Mr Venkatesh H R
Mother: Ms Shylaja K R
Marital Status: Married
Spouse: Ms Shivanee H

Nationality: Indian
Religion: Hindu
Caste: Brahmin
Sub-Caste: Smartha
Category: General

RESEARCH

Research Publications: **73**
Citations: **1980**
h-index: **22**
i-10 index: **41**
Top 2% Influential Scientist: **2024 and 2025**





RESEARCH INTERESTS

My research interests lie at the intersection of mathematical theory, computational science, and data-driven modelling, with a strong emphasis on addressing contemporary scientific and industrial challenges. In recent years, my research has actively expanded towards integrating machine learning frameworks such as Physics-Informed Neural Networks (PINNs) and Artificial Neural Networks (ANNs) with classical governing equations to enhance the analysis, prediction, and control of complex fluid flow and heat transfer problems. I am interested in hybrid modelling approaches that combine first-principles mathematics with data-driven intelligence to overcome limitations of purely analytical or numerical methods.

I also engage with computational fluid dynamics (CFD) as both a research and visualization tool, employing platforms such as ANSYS Fluent to study flow behaviour, validate mathematical models, and generate high-fidelity visual insights for complex transport phenomena. My research interests are in computational fluid dynamics with applications to thermal systems and transient flow problems relevant to energy and industrial engineering. Recently, I have conducted CFD investigations on shell-and-tube heat exchangers (STHX) to analyze flow behavior, thermal performance, and design efficiency under realistic operating conditions. In addition, I have studied sloshing dynamics in storage tanks, focusing on transient free-surface flows and their implications for structural integrity and operational safety. These studies motivate my pursuit of postdoctoral research in CFD-based analysis and optimization of engineering systems.

ADMINISTRATION

Coordinator, Centre for Mathematical Needs — **2025 - Present**
Department of Mathematics, CHRIST (Deemed to be University) | Bengaluru, India

Coordinator, School Timetable Committee — **2024 - Present**
School of Sciences, CHRIST (Deemed to be University) | Bengaluru, India

Coordinator, Internal Quality Assurance Cell — **2024 - Present**
Department of Mathematics, CHRIST (Deemed to be University) | Bengaluru, India

Coordinator, Department Timetable Committee — **2024 - 2025**
Department of Mathematics, CHRIST (Deemed to be University) | Bengaluru, India

Coordinator, Strategic Plan — **2023 - Present**
Department of Mathematics, CHRIST (Deemed to be University) | Bengaluru, India

Coordinator, SDG Cell — **2022 - 2023**
Department of Computational Sciences, CHRIST (Deemed to be University) | Delhi

Coordinator, Internal Quality Assurance Cell

I have extensive hands-on experience in academic quality assurance and institutional documentation through my role as IQAC Coordinator. My responsibilities include preparation and maintenance of IQAC documentation, support for NAAC-aligned processes, academic audits, outcome-based education (OBE) documentation, program-wise data compilation, and compliance reporting. I have actively contributed to the structuring of course files, attainment reports, faculty activity records, research metrics documentation, and continuous quality improvement (CQI) practices. My work emphasizes accuracy, traceability, and audit-readiness, supporting department in maintaining robust academic governance system.

Coordinator, Strategic Plan

I have served as the Strategic Plan Coordinator for the Department of Mathematics, actively contributing to the formulation, implementation, and monitoring of the department's three-year strategic plan aligned with the university's institutional vision and growth targets. My role involved translating university-level objectives into measurable departmental goals across research output, teaching quality, faculty development, student engagement, and academic visibility. I coordinated data collection, benchmarking, action-plan documentation, progress tracking, and outcome reporting, working closely with faculty members and academic leadership to ensure timely achievement of milestones. This experience strengthened my ability to align academic planning with institutional priorities, support evidence-based decision-making, and contribute to sustainable, goal-oriented academic development.

Coordinator, SDG Cell

During my tenure at the Delhi NCR campus, I served as the SDG Cell Coordinator, contributing to the institutional implementation of the United Nations Sustainable Development Goals through academic, research, and outreach initiatives. My role involved coordinating SDG-focused academic activities, promoting sustainability-aware teaching practices, and encouraging research aligned with themes such as sustainable energy systems, resource optimization, and environmental modelling. I facilitated student-centric activities, interdisciplinary discussions, and awareness programs that embedded sustainability perspectives into mathematics and engineering education. I also supported documentation and reporting of SDG-related initiatives for institutional records and quality frameworks, ensuring alignment with sustainability benchmarks.

EDITORIAL BOARD

Assistant Editor
CU Journal of Non-Linear Fluid Mechanics

Assistant Editor
Mapana, Journal of Sciences

TEACHING (4Y)

Visiting Lecturer **2026 – present**
Universiti Malaysia Terengganu | Terengganu, Malaysia

Assistant Professor - 6M **2025 – present**
CHRIST Online | CHRIST (Deemed to be University) | India

Assistant Professor - 2Y 6M **2023 – present**
CHRIST (Deemed to be University) | Bangalore, India

Assistant Professor - 1Y 6M **2022 - 2023**
CHRIST (Deemed to be University) | Delhi NCR, India

My teaching approach is strongly learner-centric, emphasizing conceptual clarity, active learning, and real-world application. I design courses that move beyond procedural mathematics, encouraging students to explore problem formulation, modelling, interpretation, and computational validation. I actively incorporate flipped classrooms, problem-based learning, guided discovery, and computational laboratories to accommodate diverse learning styles. I integrate digital and computational tools such as Python, Mathematica, and simulation platforms to promote exploratory learning and independent thinking. Continuous assessment, reflective assignments, and research-oriented mini-projects form a key part of my pedagogy, ensuring alignment with outcome-based education and lifelong learning objectives.

I actively engage with the use of educational technology and artificial intelligence to enhance learning experiences and academic outcomes. With a specialization in Generative AI for Educators, I integrate AI-driven tools to support personalized learning, formative assessment, and conceptual understanding while maintaining academic integrity and ethical use. I employ generative AI platforms for adaptive content creation, guided problem-solving, concept visualization, and feedback generation, enabling students to learn at their own pace and depth. My approach positions AI not as a replacement for pedagogy, but as a cognitive partner that enhances critical thinking, creativity, and learner autonomy, preparing students for AI-enabled professional and research environments.

RESEARCH TEAM

PhD Scholars - 03 (on going)

1. Ms Sini Katharin Antony
2. Ms Clair Tom
3. Ms Ananya V Kamath

PG Students - 04

1. Ms Veena R (On going)
2. Ms Shivani A (On going)
3. Ms Ananya V Kamath (Completed)
4. Ms Shivani B K (Completed)

UG Students - 05 (Ongoing)

1. Mr Venu Vamsi
2. Ms Devanshi Tuwani
3. Ms Nancy Surana
4. Ms Nehal
5. Ms Sahana Rajkumar

ACADEMIC EVENTS

Moderator **2025 Onwards**
Meet the Expert Session

Core Committee - International Conference **2026**
Scientific Advances in Natural and Theoretical Sciences - SAINTS 2026

Core Committee - National Conference **2025**
Energizing the Future: Navigating Sustainability and Climate Change

Core Committee - International Conference **2024**
69th Congress of the Indian Society of Theoretical and Applied Mathematics - ISTAM

Core Committee - International Workshop **2024**
Instabilities in Newtonian Fluid Flows

Core Committee - International Conference **2023**
Emerging Trends in Mathematical Sciences

INTERNS

Ms Sirisha B Reddy (Completed)
CHRIST (Deemed to be University), India

Mr Pavan N D (Completed)
CHRIST (Deemed to be University), India

Mr Anirudh (Ongoing)
University of Oxford, UK

LANGUAGE

- English
- Kannada
- Telugu

REFERENCES

Dr Pradeep G Siddheshwar
Senior Professor,
Department of Mathematics
CHRIST (Deemed to be University), India
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Dr Shalela Binti Mohd
Associate Professor
Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu, Malaysia
Phone: +60 11-26089528
Email : shalela@umt.edu.my

Dr Jayalatha G
Professor & Head
Department of Mathematics
RV College of Engineering, India
Phone: +91-9880693238
Email : jayalathag@rvce.edu.in

CONSULTANCY

ATOMALLOYS **2026**
Technical Consultant for Evaluation of Computational Fluid Dynamics (CFD) for 5000L tank

FUNDS RECEIVED

INR 2,00,000 for VAANI Conference - 2025 **2025**
All India Council for Technical Education

INR 2,00,000 for ISTAM-2024 **2024**
Anusandhan National Research Foundation (ANRF)

INR 5,00,000 for ISTAM-2024 **2024**
Ministry of Jalshkathi, Department of Water Resources and Ganga Rejuvenation

PROJECTS

The Study of Newtonian Fluid Flows Subject to Non-Linear Boundary Conditions

INR 3,00,000

SEED Money Project: 2024 - 2025

INVITED TALKS

Integrating Technology in Mathematics Pedagogy **2025**
Hands-on workshop for the Mathematics Teachers of Grade XI and XII
Department of Mathematics, Christ junior College, Bengaluru 560029, Karnataka, India

FOSS to Tackle Mathematical Problems **2025**
FDP under the RUSA Project
Department of Mathematics, University of Mysuru, Mysuru, Karnataka

Roadmap for Research **2024**
Department of Mathematics, St. Claret College, Bengaluru 560087, Karnataka, India

ONLINE COURSES

1. **Generative AI: Empowering Modern Education | IBM**
2. **Indigenous Religions & Ecology | Yale University**
3. **Ethical AI: AI essentials for everyone | University of Cambridge**
4. **The Talmud: A Methodological Introduction | Northwestern University**
5. **Generative AI: Introduction and Applications | IBM**
6. **AI in Education: Leveraging ChatGPT for Teaching | University of Pennsylvania**
7. **Generative AI: Prompt Engineering Basics | IBM**
8. **Aboriginal Worldviews and Education | University of Toronto**
9. **The Ancient Greeks | Wesleyan University**

THANK YOU

TEACHING STATEMENT

Philosophy and Context

My academic practice is grounded in the belief that rigorous mathematics, when communicated with clarity and relevance, can significantly shape the intellectual confidence and problem-solving abilities of engineering students in India. As a mathematician working at the interface of theory, computation, and application, I view teaching and research as mutually reinforcing activities. Research keeps my teaching contemporary and inquiry-driven, while classroom engagement constantly reshapes the questions I ask as a researcher.

Indian classrooms are uniquely diverse academically, linguistically, and socio-economically. Teaching in such an ecosystem demands not only subject expertise, but also sensitivity, structure, and adaptability. My approach to teaching mathematics and computation is therefore learner-centric, scaffolded, and strongly aligned with the cognitive and professional needs of Indian engineering students.

Writing as a Tool for Conceptual Retention

One of the most powerful yet underutilized tools in mathematics education is structured writing. I actively encourage students to write mathematics not merely solve problems. Through reflective summaries, derivation notebooks, error-analysis reports, and concept-explanation assignments, students are guided to articulate why a method works, not just how it is applied.

Writing forces students to slow down their thinking, organize ideas, and confront gaps in understanding. I have observed that students who regularly write short conceptual explanations demonstrate stronger long-term retention and transfer of knowledge, especially in courses such as numerical methods, fluid mechanics, and mathematical modelling. This practice is particularly effective in the Indian context, where examination-oriented learning often suppresses conceptual articulation. Writing restores ownership of ideas and transforms passive learners into active thinkers.

The Need for Rigorous and Structured Tutorials

In my teaching practice, tutorials are not treated as remedial sessions but as core learning spaces for Gen Z learners. I design tutorials to be rigorous, interactive, and progressively challenging, focusing on problem decomposition, estimation, algorithmic thinking, and interpretation of results. Each tutorial is structured to bridge the gap between lecture material and real-world application.

For engineering students many of whom are transitioning from formula-driven school mathematics, tutorials serve as the arena where confidence is built. I emphasize guided problem-solving, peer discussion, and multiple-solution strategies. Rigorous tutorials cultivate mathematical maturity, reduce fear of abstraction, and prepare students for open-ended engineering problems rather than closed-form textbook questions.

Understanding Student Needs

A central element of my pedagogy is diagnosing student needs early and continuously. Through diagnostic tests, classroom questioning, informal feedback, and reflective assignments, I assess students' preparedness, misconceptions, and learning barriers. This allows me to adapt pacing, instructional methods, and assessment strategies in real time. Indian classrooms often contain students with vastly different academic backgrounds. Recognizing this, I employ differentiated instruction offering foundational reinforcement where needed, while simultaneously challenging advanced learners through exploratory problems and mini-projects. Teaching, in my view, is effective only when it is responsive.

I place strong emphasis on supporting both slow and fast learners with sensitivity and confidentiality. Slow learners are assisted through one-to-one mentoring, scaffolded problem sets, and confidence-building interventions without public labeling or academic stigma. Fast learners are provided with enrichment tasks, research-oriented readings, and advanced problem challenges in a discreet manner that preserves classroom equity and peer dynamics. This balanced approach ensures inclusivity while nurturing academic excellence.

Curating and Propagating Quality Resources

In an era of information overload, the role of a teacher extends to that of a curator of quality knowledge. I actively collect, evaluate, and disseminate high-quality learning resources, including textbooks, lecture notes, open-source computational tools, simulations, and research-based articles. Students are guided on how to read technical material, what to focus on, and how to connect resources meaningfully.

By introducing students to structured repositories of learning rather than random online searches, I help them develop disciplined self-learning habits. Alongside traditional resources, I promote the judicious and ethical use of AI-powered learning tools such as ChatGPT, Gemini, and Perplexity. These tools are introduced as cognitive aids for idea clarification, conceptual reinforcement, exploratory questioning, and reference discovery—rather than as shortcuts for answers. Students are trained to critically evaluate AI-generated content, verify results mathematically, and use these platforms to enhance reasoning, not replace it.

A Differentiated Approach

Teaching in India requires conscious differentiation between rural and urban student cohorts, without compromising academic standards. Rural students often possess strong perseverance and intuitive reasoning but may lack early exposure to computational tools, academic English, or exploratory learning environments. For such students, I emphasize confidence-building, gradual abstraction, language-sensitive explanations, and hands-on computation.

Urban students, while often more exposed to technology and competitive environments, may exhibit gaps in conceptual depth and patience with rigor. With them, I emphasize disciplined reasoning, derivations, proof-based thinking, and reflective learning. In both cases, the goal is convergence bringing all students to a common platform of mathematical confidence, autonomy, and professional readiness.